HO CHI MINH STOCK EXCHANGE

Interface Specification for Market Data Distribution System

Version 1.6

2023.08.01

Release/Revision history

No	Version	Date	Release/Revision Description	Author
1	1.0	2018.01.02	New	HOSE
2	1.1	2018.04.05	1. Modify the name and fields of the message 'Reference Price Determination(M7)' to 'Security Information Notification(M7)' 2. Add the message 'Price Limit Expansion(MX)' in interface list. 3. Delete the message 'Dynamic Price Limit(MG)', 'Issuer Information(MJ)' and 'Member(Dealer) Information(MK)' Security Board (M6). 4. Merge the definition of the messages Price + Price(Bond), Deem Trade Price + Deem Trade Price(Bond). 5. Modify field meaning and length of DisparateRatio(30602) to merge with DisparateSign(30601), and delete DisparateSign(30601). 6. Remove ForeignerLimitRatio(30598) from the message 'Foreigner Order Limit(MF)' 7. Delete the field HaltReason(327) from the definition of Security Status. 8. Add a new tag DynamicVIBasePrice(20047) and DynamicVIDispartiyRatio(20048) in 'Volatility Interruption'. 9. Add new Tag 30606 “TotalMsgNo” and Tag 30607 “CurrentMsgNo” in ‘Index Constituents Information’ 10. Edit the tag MarketMakerContractCode(30574) in 'Market Maker Information' 11. Add TickerCode(30624), SymbolShortCode(30628), SymbolName(30629), SymbolEnglishName(30630), SymbolCloseInfoPxType(20027), VWAP(30625), ProductID(20009), ProductGrpID(20003), ListedShares(20020), SecurityGroupID(30604), ContractSize(30631), in the definition of 'SecurityDefinition'. 12. Remove SecurityDesc(107), SecurityGroup(1151), SecurityType(167), UnitOfMeasureQty(1147), MarginBasePrice(30571), SettlementPriceTypeCode(30572), ExClassType(20018) in the definition of 'SecurityDefinition'. 13. Delete SecurityIDSource(22) in all the message. 14. Update definition ‘Format” and “Length” of many of tags in the document. 15. Modify field meaning and possible value of many tags in the document. 16. Updated list of Messages.	HOSE
3	1.2	2018.07.18	1. Modify the field meaning of TotNumReports(911) to more general meaning, and the length to 16 2. Modify all the statistics messages to split so that their messages does not include repeat groups; (top N, number of invest type codes) 3. Add TotNumReports(911) to the definitions of 'Top N' statistics 4. Modify ForeignInvestorTypeCode(30581) to ForeignInvestTypeCode(20054) in the definition of 'Trading Result of Foreign Investors(MT)' 5. Modify the length of BuyVolume(330) and SellVolume(331) to 10 for synchronization to meta definition 6. Modify DayCumQty(425) to TotalVolumeTraded(387) in the definition of 'Index(M1)' 7. Modify ClsPx(30564) to SymbolCloseInfoPx(20026) in the definition of 'Price Recovery(W)'	HOSE

No	Version	Date	Release/Revision Description	Author
			8. Modify 'Symbol Closing Information(M8)' to Batch in the 3. List of Messages 9. Add new tags and fields about continutous auction and block trading volumes/values to 'Index(M1)' 10. Delete BaseCurrencyIndexYN(30570) from 'Index(M1)' and 'Investor per Industry(M2)' and 'Index Constituents Information(ML)' 11. Add FirstTradingDate(30642) to the definition of 'Security Definition(d)' 12. Modify the definition of tag FornInvestTypeCode (20054). 13. Modify the length of MDEntrySize(271) to synchronize to meta system 14. Delete BordID(20054) in the definition of 'Investor per Industry(M2)', 'Investor per Symbol(M3)', 'Top N Members per Symbol(M4)', 'Top N Symbols(MP, MQ, MR, MS)'; those statistics merges not per {Symbol, Board} but per {Symbol} 15. Modify the field meaning of IndexsTypeCode(30167) 16. Add ForeignerBuyPosblQty(30557) to the definitions of 'Foreigner Order Limit'. 17. Add four tags about trading volumes and amounts to the definition of 'Trading Result of Foreign Investors(MT)'.	
4	1.3	2019.08.23	1. Add new tag EvaluationPrice(20014), HgstOrderPrice(20015), LwstOrderPrice(20016), and add it to the definition of 'Security Definition(d)' and 'Security Information Notification(M7)' 2. Modify Req'd fields in 'Security Definition'(d)' 3. Delete BoardID(20004) from the definition of 'Foreign Order Limit(MF)' 4. Delete TransactTime(60) from the definition of 'Investor per Symbol(M3)' 5. Modify the length of ForeignerOrderLimitQty(30558) from 15 to 16 6. Add BoardEvtID(20005) to the definition of 'Security Status(f)', and modify Req'd of SessOpenCloseCode(20008) in the definition of 'Security Status(f)' 7. Modify the length of SellVolume(Total)(30643), BuyVolume(Total)(30645) from 10 to 12 for synchronization to meta definition 8. Modify Req'd fields in 'Disclosure(MU)' 9. Add possible values for Currency(15), SecurityExchange(207), TradingSessionID(336), ProductGrpID(20003),BoardID(20004), BoardEvtId(20005) and MarketID(30001) 10. Add market HCX to the definition of MarketID(30001) and ProductGrpID(20003) 11. Modify possible values for SymbolTradingMethodStatusCode(30636)	
5	1.4	2020.05.29	1. Modify the field description of BoardID(20004)	HOSE
6	1.5	2022.08.05	1. Modify the length of BuyVolume(330) and SellVolume(331) from 10 to 12 due to the possibility of overflow 2. Modify MarketIndexClass(30569) 3. Add new tag Sector Type Code(30647) and add it to the definition of 'Security Definition(d)' 4. Modify the length of DisclosureID(30605) from 6 to 16	HOSE

No	Version	Date	Release/Revision Description	Author
			5. Delete the possible valule(5 = Closing Price) in MDEntryType(269) 6. Modify Req'd fields Body in 'Disclosure(MU)' 7. Modify Req'd field RandomEndReleaseTime in 'Random End(MW)' 8. Add 'Timming' in the table of List Messages. 9. Modify Req'd field TradingSessionID in Security Status(f) 10. Modify Req'd field Symbol in Security Status(f)' 11. Modify possible values for BoardEvtID(20005)	
7	1.6	2023.08.01	1. Add TscProdGrpId(30648) in to the definition of 'Security Status (f)'. 2. Delete OpenInterestQty(30540) from the definition of 'Security Definition (d)' 3. Delete SettlementPrice(30573) from the definition of 'Security Definition (d)' 4. Add new tag SettlementPrice(30573) and add it to the definition of 'Open Interest(MA) 5. Add PleUpLmtStep(30649) and PleLwLmtStep(30650) to the definition of 'Price Limit Expansion (MX)'	

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1 Introduction

1.1 Objective

This document describes messages in format of FIX Protocol which are used by Market Data Distribution System at HOSE to provide market information to outside organizations. Only authorized messages have been described in this document. Every message describes required fields, value and length of fields.

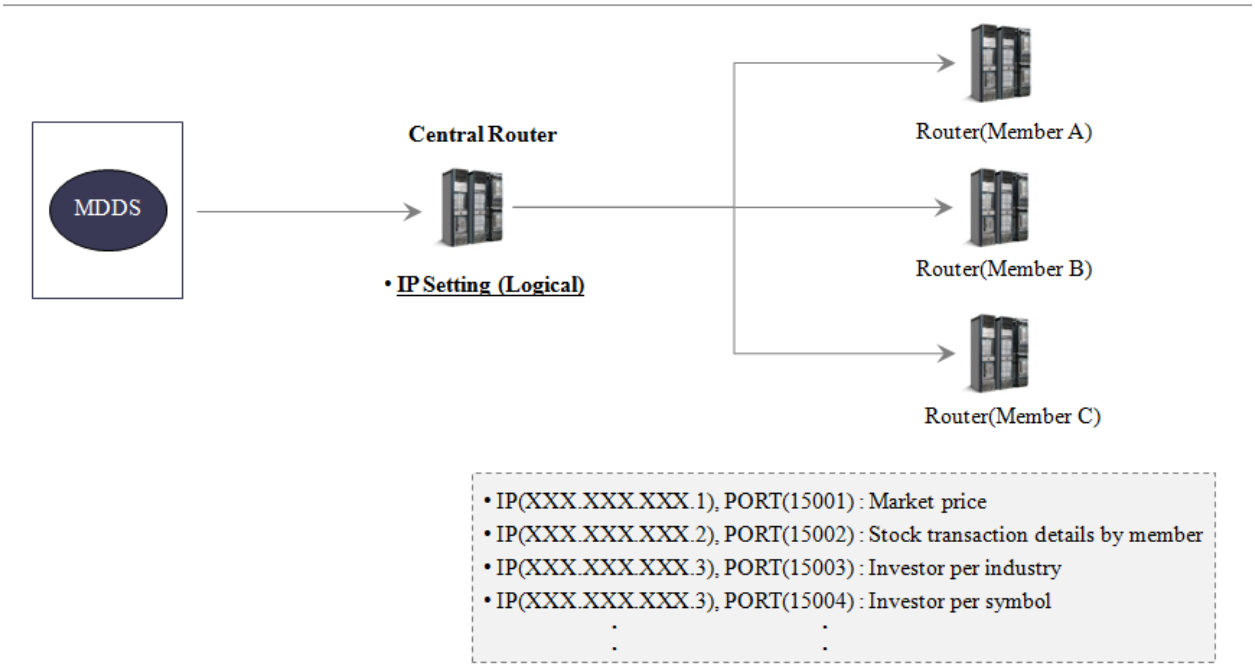
1.2 Scope of document

Market Data Distribution System provides market information such as: Best orders; indices; quantity of trading; value of trading … The Market Data Distribution System also provides information which is relates to disclosure and corporation action of listed company.
The information which relates to information of orders, result of orders… will not distributed by this system.

2 Market data distribution

2.1 Message Distribution

Market data of all securities traded in the Exchange shall be distributed to members and vendors via leased lines. UDP multicast method provides all market data subscribers with the same data at the same time.

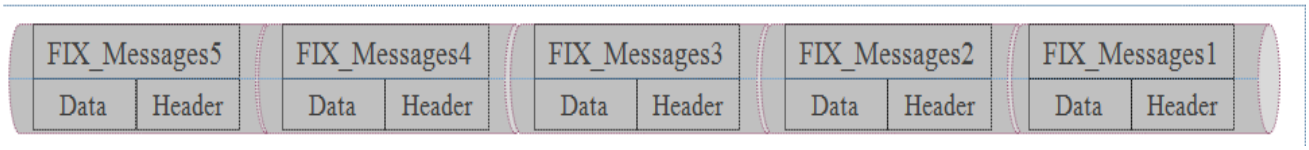


2.2 Data Category and Frequency of data distribution

Data Category and Frequency of data distribution.

Data Category		Example Data	Frequency of data distribution	Recovery Mechanism
Real time data		● Price (deal information, levels of best order) etc.	● During trading hours (e.g. 09:00 ~ 15:00)	**Sending x messages per second(parameter)**
Periodic data		● Index, price recovery top N members per symbol, investor per industry, ETF iNav etc.	● Particular time interval during trading hours (e.g.10 seconds, 30 seconds	
Batch Data	**EOD data**	● Closing price, closing index, Investor per symbol etc	● After the market closed (e.g 15:00~ 18:00)	
	Batch data	● Security definition, security status, symbol event,issuer information etc	● Before the markets opens (e.g 06:00~07:30)	

Each message is made up by "Header part + Data part"



2.3 Supplementary Function

- **Periodic Recovery**

Best orders and current price of each symbol are distributed in multicast periodically during trading time (e.g. 09:00~15:00). (More detail: *Price Recovery* message).

- **Manual Recovery**

Market Data Subscribers may request for re-transmission some parts of distributed messages (in case of data loss or omission). Requested information shall be re-sent to Receiver using unicast method during trading time and/or after the market closed. The receiver should request a range of sequence or time to exchange system operator.

- **Polling Check**

Polling is used to check network condition between information system and information use.
Interval is 1 minute (example) but adjustable, if necessary.
Polling check is transmitted to check the connectivity 24 hours a day, 365 days a year.

- **Possible Duplicates**

Distributed data can be duplicated but must not be omitted.
If information users receive the duplicate data, it is their responsibility to handle the message. (Member has to make some logic to skip of it when receiving a duplicated message)
Receivers should make some logic to check and ignore duplicated messages.

3 List of Messages

No	Interface Name	MsgType	Timing	Online /Batch	Purpose
1	Security Definition	d	DO	Batch	Phân phối dữ liệu tham chiếu để nhận dữ liệu thị trường để bảo mật
2	Security Status	f	ONT	Online	Distribute market control to change the status of a security

3	Security Information Notification	M7	ONT	Online	Distribute determination/notification of reference price, valuation price, or list of shares among market control
4	Symbol Closing Information	M8	MC	Batch	Distribute symbol closing information among market control
5	Volatility Interruption	MD	ONT	Online	Distribute volatility interruption message among market control
6	Market Maker Information	MH	DO	Batch	Distribute market maker information where the unit of contracts is symbol for stock/bond market and is product for derivatives
7	Symbol Event	MI	DO	Batch	Distribute symbol event for each symbol and the reason of the event for Stock market
8	Index Constituents Information	ML	DO	Batch	Distribute the list of constituent securities for each index
9	Random End	MW	ONT	Online	Distribute the information of random end when the random end occurs by conditional trigger
10	Price	X	ONT	Online	Distribute the repeated structures of best orders in levels for best order messages, and distribute repeated structures of the execution/opening/highest/lowest price for trade messages
11	Price Recovery	W	ONT	Online	Distribute price recovery periodically (automatically)
12	Price Recovery(Bond)	W	ONT	Online	Distribute price recovery with yield periodically (automatically)
13	Index	M1	ONT	Batch/ Online	Distribute index periodically during trading time and EOD(from the end of the market to termination of process) The indices which are calculated only in EOD are distributed by batch process
14	Investor per Industry	M2	ONT	Online	Distribute sum of trade by investor for each industry
15	Investor per Industry(Bond)	M2	ONT	Online	Distribute sum of trade by investor for each bond category
16	Investor per Industry(Derivatives)	M2	ONT	Online	Distribute sum of trade by investor for each future/option market
17	Investor per Symbol	M3	MC	Batch	Distribute sum of trade by investor for each symbol
18	Top N Members per Symbol	M4	ONB	Online	Distribute top N members in the structures repeated for each symbol
19	Open Interest	MA	DB	Batch	Distribute open interest received from VSD in EOD
20	Deem Trade Price	ME	ONT	Online	Distribute deem trade price with yield for pre-opening session
21	Foreigner Order Limit	MF	ONO	Batch/ Online	Distribute the limited quantity to order by foreigners which is changed by orders and executions for each symbol
22	Price Limit Expansion	MX	ONT	Online	Distribute expansion information of price limit when occurred by market control
23	ETF iNav	MM	ONT	Online	Distribute iNav calculated periodically for each ETF symbol
24	ETF iIndex	MN	ONT	Online	Distribute iIndex calculated periodically for each ETF symbol
25	ETF TrackingError	MO	DO	Batch	Distribute tracking error calculated for each ETF symbol
26	Top N Symbols with Trading Quantity	MP	ONT	Online	Distribute top N symbols with trading quantity in the structures repeated N times for each market
27	Top N Symbols with Current Price	MQ	ONT	Online	Distribute top N symbols with current price in the structures repeated N times for each market

28	Top N Symbols with High Ratio of Price	MR	ONT	Online	Distribute top N symbols with high ratio of price in the structures repeated N times for each market
29	Top N Symbols with Low Ratio of Price	MS	ONT	Online	Distribute top N symbols with low ratio of price in the structures repeated N times for each market
30	Trading Result of Foreign Investors	MT	ONT	Online	Distribute sum of trade by foreign investor for each type code of foreign investors
31	Disclosure	MU	ONB	Batch/ Online	Distribute disclosure received from legacy disclosure system and MDIS
32	Time Stamp (Polling)	MV	ALL	Online	Distribute time stamp to check polling for each physical network line

Timing code of message communication
- DO: Day Opening (Before or At Market Opening)
- ONO: Online (From Day Opening until Market Closing)
- ONT: Online (Trading Hours)
- ONB: Online (From Market Opening until Daily Batch)
- ALL: All Day
- MC: Market Closing
- DB: Daily Batch
- DC: Day Closing (EOD, End of Day)

4 Detailed messages

- **Header**

Tag	Field Name	Req'd	Format	Length	Remarks
8	BeginString	Y	String	7	FIX.4.4
9	BodyLength	Y	Int	6	Length filed
35	MsgType	Y	String	3	**Specification of MsyType(35)**
49	SenderCompID	Y	String	16	Sender
56	TargetCompID	Y	String	16	99999
34	MsgSeqNum	Y	Int	11	Session sequence number
52	SendingTime	Y	UTCTime	21	Time of message transmission

*** Specification of MsyType(35)**

d = Security Definition
f = Security Status
X = Price
W = Price Recovery
M1 = Index
M2 = Investor per Industry
M3 = Investor per Symbol
M4 = Top N Members per Symbol
M7 = Security Information Notification
M8 = Symbol Closing Information
MA = Open Interest
MD = Volatility Interruption
ME = Deem Trade Price
MF = Foreigner Order Limit
MH = Market Maker Information
MI = Symbol Event
MX = Price Limit Expansion
MW = Random End
ML = Index Constituents Information
MM = ETF iNav

MN = ETF iIndex
MO = ETF TrackingError
MP = Top N Symbols with Trading Quantity
MQ = Top N Symbols in Current Price
MR = Top N Symbols with High Ratio of Price
MS = Top N Symbols with Low Ratio of Price
MT = Trading Result of Foreign Investors
MU = Disclosure
MV = Time Stamp (Polling)

- **Trailer**

Tag	Field Name	Req'd	Format	Length	Remarks
10	CheckSum	Y	String	3	Three byte, simple checksum

4.1 Security Definition

This message contains all the reference data for a security and is distributed before market open.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = d
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
20004	BoardID	Y	String	2	Unique identifier to a Board where a given instrument is traded G1 : Main G2 : Pre open G3 : Post closing G4 : Odd lot G7 : Buy-in G8 : Sell-out T1 : Negotiation (regular) T4 : Negotiation (regular for Odd lot) T2 : Negotiation (pre) T3 : Negotiation (post) T6 : Negotiation (post for Odd lot) R1 : Negotiation (Repo) AL : All Boards
911	TotNumReports	Y	Int	16	Entire number of messages for issue information transmission or for a set of statistics results
207	SecurityExchange	Y	String	2	The unique ID assigned by Exchange HO: HoSE HX: HNX
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want

30624	TickerCode	Y	String	20	Symbolic characters of an instrument. It might be used as ticker code (a.k.a. trading code, local code, stock code, bond code and so forth)
30628	SymbolShortCode	Y	String	9	A short unique code for an instrument which is determined at exchange and used for more convenience than 12-digit instrument code (ISIN code). It may be used as a part of Repo instrument's ISIN code
30629	SymbolName	Y	String	120	Instrument name
30630	SymbolEnglishName	Y	String	80	Instrument English name
20009	ProductID	Y	String	11	Product is a layer or unit to define or manage rules of trading, clearing and settlement, instrument creating, calculating statistics, etc. Product ID is assigned to Product ID created by combination of exchange, market, security group, underlying asset and currency List of Products (ID and description) will be defined later.
20003	ProductGrpID	Y	String	3	Indicates Product Group ID which is defined in the trading system. Product Group is a group of Products that share the trading schedule. STO : Stock Market Product Group BDO : Bond Market Product Group RPO : Repo Market Prodcut Group STX : Listed Stock Market Product Group UPX : UPCOM Product Group BDX : Government Bond Product Group FIO : Index Future Product Group FBX : Bond Future Product Group HCX : Corporate Bond Product Group
30604	SecurityGroupID	Y	String	2	Security group ID (Stock, Investment Trust, Futures, Option, etc.)
201	PutOrCall	N	String	1	Rights type code of derivatives P = Put C = Call
1194	ExerciseStyle	N	String	1	Rights exercise type code of a derivatives A: American-style E: European-style B: Bermuda-style Z: Etc.
200	MaturityMonthYear	N	String	6	Expiration or maturity month (YYYYMM)
541	MaturityDate	N	String	8	Expiration or maturity date YYYYMMDD
106	Issuer	N	String	10	Issuer's unique ID
225	IssueDate	N	String	8	Issued date YYYYMMDD
231	ContractMultiplier	Y	Float	22(12.8)	Trading multiplier for calculating trading or settlement amount
223	CouponRate	N	Float	14(7.5)	Rate of return of a bond. Coupon Rate = Annual Interest Payments / Par Value
15	Currency	Y	String	3	Currency code on ISO 4217 VND: Vietnamese Dong USD: US Dollar EUR: Euro
20020	ListedShares	N	Int	16	Number of listed stocks
1149	HighLimitPrice	Y	Price	15(9.4)	Available high limit price for the tradig day
1148	LowLimitPrice	Y	Price	15(9.4)	Available low limit price for the tradig day
202	StrikePrice	Y	Price	18(9.8)	Exercise (strike) price of options or Subscription Warrant

965	SecurityStatus	Y	String	1	Whether trading halt or not Y = Halt N = Not Halt
30631	ContractSize	Y	Float	21(12.8)	The quantity of trading objects (in case of derivatives, underlying assets) per unit trade or contract. For example, 500 point for index, 100 shares for stock, 100,000 amount for bond or foreign currency, 10kg for gold. In case of derivatives it is the basis of physical delivery
1193	SettlMethod	N	String	1	The final settlement method C: Cash D: Physical Delivery A: Cash + Physical Delivery O: None
236	Yield	N	Float	13(5.6)	Yield to maturity based on Reference Price
20013	ReferencePrice	Y	Price	15(9.4)	Reference price of a given instrument. Price limit is determined based on this price.
20014	EvaluationPrice	N	Price	15(9.4)	Valuation price for a 'Re-Valuation' instrument. For such instrument, price limit is set based on this price until the first execution. Once the first execution is occurred, price limit is set again base on the price of the first execution.
20015	HgstOrderPrice	N	Price	15 (9.4)	Upper price limit for a 'Re-Valuation' instrument during the opening call auction (or until the first execution).
20016	LwstOrderPrice	N	Price	15 (9.4)	Lower price limit for a 'Re-Valuation' instrument during the opening call auction (or until the first execution).
140	PrevClosePx	Y	Price	15(9.4)	Prev Close Price
20027	SymbolCloseInfoPxType	Y	String	1	Type of closing price 1: Actual Trade 2: Biased Quotation 3: No Trade 4: Biased Quotation of an Re-Valuation Instr.
30642	FirstTradingDate	N	String	8	First trading date YYYYMMDD
30511	FinalTradeDate	N	String	8	Last trading date YYYYMMDD
30512	FinalSettleDate	N	String	8	Last settlement date YYYYMMDD
30301	ListingDate	N	String	8	Listing date of a derivative product or an instrument YYYYMMDD
30614	RandomEndTriggeringConditionCode	Y	String	1	Conditions for triggering random-end at the end of every call auction session. In case of code '2' (Conditional Triggering) the actual triggering is conducted after checking if price conditions meet the 'Random End Rule' 0: Not Trigger 1: Unconditional Trigger 2: Conditional Trigger
20018	ExClassType	N	String	2	Corporate action type that may affect reference price. 00: N/A 01: Stock Dividend/Bonus (D/B) 02: Cash Dividend 04: Right Offering (RO) 03: Stock D/B + Cash D 05: RO + Stock D/B 06: RO + Cash D 07: RO + Stock D/B + Cash D 08: Convertible Bond

					16: Treasury Stock Div. P1: Stock Split P2: Stock Merge XX: Etc.
30625	VWAP	Y	Float	15(9.4)	Volume weighted average price
30635	SymbolAdminStatusCode	N	String	3	Codes to identify administrative status of a symbol due to breaching exchange regulation about public disclosure, fair trade and so on NRM: Normal WID: Warning (Information Disclosure) WFR: Warning (Financial Report) WOV: Warning (Other Violations) CTR: Controlled C&R: Controlled & Restricted RES: Restricted
30636	SymbolTradingMethodStatusCode	N	String	3	Codes to identify trading method type of a specific symbol NRM: Normal SNE: Special Trading for Non-Execution SLS: Special Trading after Long Suspension NWN: New Listing with Normal Price Limit NWE: New Listing with Exceptional Price Limit
30637	SymbolTradingSantionStatusCode	N	String	3	Codes to classify status of trading sanction such as halt (or suspension), trading forbidden for some reasons NRM: Normal DTL: Delisted for Transfer List SUS: Suspension TFR: Trading Forbidden for Restricted
30647	SectorTypeCode	N	String	2	Market segment code to classify instruments by such as credit, trading volume, market capitalization and so on ▦▦ Codes ▦▦ P: Premier S: Standard A: Alternative
	Standard Trailer	Y			

4.2 Security Status

This message provides information on the status of a market segment. It is sent whenever there is the change of trading session.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = f
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market

					- UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
30648	TscProdGrpId	Y	String	3	ID for identifying the set of the same TSC(Trading Schedule Control) target products. A logical unit controlled by the same schedule. ▦▦ IDs ▦▦ STO : HoSE Stock BDO : HoSE Bond RPO : HoSE Repo BDX : Government Bond FIO : HoSE Index Futures FBX : HNX Bond Futures STX : HNX Stock UPX : HNX UpCom HCX : HNX Corporate Bond
20004	BoardID	Y	String	2	Unique identifier to a Board where a given instrument is traded G1 : Main G2 : Pre open G3 : Post closing G4 : Odd lot G7 : Buy-in G8 : Sell-out T1 : Negotiation (regular) T4 : Negotiation (regular for Odd lot) T2 : Negotiation (pre) T3 : Negotiation (post) T6 : Negotiation (post for Odd lot) R1 : Negotiation (Repo) AL : All Boards
20005	BoardEvtID	Y	String	3	Indicates Board event ID which is defined in the trading system. Trading schedule is essentially a sequence (list) of Board events, and each event is identified by three-character code or ID. Below is an example trading schedule and its Board events (and ID in parenthesis). 0915: Opening call auction (AA1) 0930: Continuous trading (BB1) 1430: Closing call auction (BC1) 1445: Market close (AC2) AA1: Opening CA Open AB1: Cont.A Open AB2: Board Trading Close AC2: Closing CA Execution AD1: Order Reception Start AD2: CA Excecution AE1: Market Halt Release AE8: Market Halt AF1: CA Open after CB AF6: CB Notice AF7: CB Notice Clear AF8: Suspension by CB AQ3: Interface Connection Cut-off

					AT6: PLE Notice AW8: Lunch Break AW9: Open after Lunch Break AX1: Board Start BB1: Cont.A Open for Non-PCA Instr. BC1: Closing CA Open BE9: Cont.A Open after Market Halt BF9: Cont.A Open after CB CC1: Closing PCA Open CD1: PCA Open CD3: PCA Execution DB2: Trading Close for LTD Instr. DC2: Closing CA Execution for LTD Instr. EI1: Trading Halt Release by Instr./Product EI8: Instr./Product Halt EI9: Cont.A Open after Instr./Product Halt GT1: GT Order Load
20008	SessOpenCloseCode	N	String	2	Indicates whether a given Board or Session is started or ended. BS : Board Start BE : Board End SS : Session Start SE : Session End SH : Session Halt SR : Session Restart
55	Symbol	N	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
336	TradingSessionID	N	String	2	Trading Session 01 = GT Order Load 10 = Opening Call Auction 11 = Opening Call Auction Extension 20 = Call Auction after Halt 21 = Call Auction after Halt Extension 30 = Closing Call Auction 40 = Continuous Auction 50 = VI Call Auction 51 = VI Call Auction Extension 60 = Order Accept Only 80 = Periodic Call Auction 90 = Trading Halt 99 = Board Closing
	Standard Trailer	Y			

4.3 Security Information Notification

This message is generated whenever there is change or update of Reference price, lower limit price and upper limit price of a symbol by market control.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = M7
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market

					- RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
20004	BoardID	Y	String	2	Unique identifier to a Board where a given instrument is traded G1 : Main G2 : Pre open G3 : Post closing G4 : Odd lot G7 : Buy-in G8 : Sell-out T1 : Negotiation (regular) T4 : Negotiation (regular for Odd lot) T2 : Negotiation (pre) T3 : Negotiation (post) T6 : Negotiation (post for Odd lot) R1 : Negotiation (Repo) AL : All Boards
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
20013	ReferencePrice	N	Price	15(9.4)	Reference price of a given instrument. Price limit is determined based on this price.
1149	HighLimitPrice	N	Price	15(9.4)	Available high limit price for the tradig day
1148	LowLimitPrice	N	Price	15(9.4)	Available low limit price for the tradig day
20014	EvaluationPrice	N	Price	15(9.4)	Valuation price for a 'Re-Valuation' instrument. For such instrument, price limit is set based on this price until the first execution. Once the first execution is occurred, price limit is set again base on the price of the first execution.
20015	HgstOrderPrice	N	Price	15 (9.4)	Upper price limit for a 'Re-Valuation' instrument during the opening call auction (or until the first execution).
20016	LwstOrderPrice	N	Price	15 (9.4)	Lower price limit for a 'Re-Valuation' instrument during the opening call auction (or until the first execution).
20020	ListedShares	N	Int	16	Number of listed stocks
20018	ExClassType	N	String	2	Corporate action type that may affect reference price. 00: N/A 01: Stock Dividend/Bonus (D/B) 02: Cash Dividend 04: Right Offering (RO) 03: Stock D/B + Cash D 05: RO + Stock D/B 06: RO + Cash D 07: RO + Stock D/B + Cash D 08: Convertible Bond 16: Treasury Stock Div.

					P1: Stock Split P2: Stock Merge XX: Etc.
	Standard Trailer	Y			

4.4 Symbol Closing Information

This message is generated at the end of day providing the last execution price of that day for each symbol. If there is no execution it is replaced with the reference price, and if no reference price is given then it will be 0 (zero).

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = M8
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
20004	BoardID	Y	String	2	Unique identifier to a Board where a given instrument is traded G1 : Main G2 : Pre open G3 : Post closing G4 : Odd lot G7 : Buy-in G8 : Sell-out T1 : Negotiation (regular) T4 : Negotiation (regular for Odd lot) T2 : Negotiation (pre) T3 : Negotiation (post) T6 : Negotiation (post for Odd lot) R1 : Negotiation (Repo) AL : All Boards
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
20026	SymbolCloseInfoPx	Y	Price	15(9.4)	The last execution price of that day. If there is no execution it is replaced with the reference price, and if no reference price is given then it will be 0 (zero)
30541	SymbolCloseInfoYield	Y	Float	13(5.6)	Final yield of a bond calculated using the closing price of that day
20027	SymbolCloseInfoPxType	Y	String	1	Type of closing price 1: Actual Trade 2: Biased Quotation 3: No Trade 4: Biased Quotation of an Re-Valuation Instr.
	Standard Trailer	Y			

4.5 Volatility Interruption

This message is generated whenever Volatility Interruption control (VI) is activated. When VI is activated, trading method of the given symbol is changed from continuous trading to call auction.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = MD
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
20004	BoardID	Y	String	2	Unique identifier to a Board where a given instrument is traded G1 : Main G2 : Pre open G3 : Post closing G4 : Odd lot G7 : Buy-in G8 : Sell-out T1 : Negotiation (regular) T4 : Negotiation (regular for Odd lot) T2 : Negotiation (pre) T3 : Negotiation (post) T6 : Negotiation (post for Odd lot) R1 : Negotiation (Repo) AL : All Boards
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
20030	VITypeCode	Y	String	1	Indicates whether VI is activated or released. 1 = Activated 2 = Released
20031	VIKindCode	Y	String	1	Indicates which kind of VI is activated or released. 1 = Static VI 2 = Dynamic VI 3 = Static VI + Dynamic VI
20032	StaticVIBasePrice	Y	Price	15 (9.4)	Indicates static VI reference price. Price range for static VI is determined based on this price, and static VI is activated when an order is about to be executed outside the VI price range.
20047	DynamicVIBasePrice	Y	Price	15 (9.4)	Indicates dynamic VI reference price. Price range for dynamic VI is determined based on this price, and dynamic VI is activated when an order is about to be executed outside the VI price range.
20033	VIPrice	Y	Price	15 (9.4)	VI Triggering Price
20034	StaticVIDispartiyRatio	Y	Price	8(3.3)	Indicates static VI activation ratio expressed in percent. static VI price range is defined as range from static VI reference price plus/minus this ratio. For example, if static VI reference price is 1000 and this ratio is 2.5, then static VI price range.
20048	DynamicVIDispartiyRatio	Y	Price	8(3.3)	Indicates dynamic VI activation ratio expressed in percent. Dynamic VI price range is defined as range from Dynamic VI reference price plus/minus this ratio. For example, if Dynamic VI reference price is

					1000 and this ratio is 2, then Dynamic VI price range is from 980 to 1020.
	Standard Trailer	Y			

4.6 Market Maker Information

Market Maker message is generated at the start of trading day for securities that have Market Maker.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = MH
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
30574	MarketMakerContractCode	Y	String	12	Unit of agreement objects for market making 1) Stock/Bond Market: Instrument ISIN code 2) Derivatives Market: Product ID
20023	MemberNo	Y	String	5	ID to identify or distinguish market participants or members with 5-digit numeric characters
	Standard Trailer	Y			

4.7 Symbol Event

This message is generated whenever there is change of the status of a security by market control

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = MI
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
30575	EventKindCode	Y	String	2	Event code for such as trading halt, low liquidation and so forth. Code '91' (Trading Halt for Repo) is only used in Gateway Repo trading system, not the Terminal Repo trading system 01: Trading Halt 04: Liquidation Trading 91: Trading Halt for Repo
30576	EventOccurrenceReasonCode	Y	String	4	Reason for designating a specific event (trading halt, administrative instrument, etc.) to an instrument or a product

30577	EventStartDate	Y	LocalMktDate	8	Starting date of an event
30578	EventEndDate	Y	LocalMktDate	8	Ending date of an event
	Standard Trailer	Y			

4.8 Index Constituents Information

Distribute the list of constituent securities for each index

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = ML
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
30569	MarketIndexClass	Y	String	1	Market code of an index 1: HoSE 2: HNX 3: Composite (HoSE + HNX) 4. UPX
30167	IndexsTypeCode	Y	String	8	Industry Index Code for example 001: Composite index 002: VNALLshare index 100: VN30 index
15	Currency	Y	String	3	Currency code on ISO 4217 VND: Vietnamese Dong USD: US Dollar EUR: Euro
30632	IdxName	Y	String	120	Index Name
30633	IdxEnglishName	Y	String	80	Index English Name
30606	TotalMsgNo	Y	Int	16	1. Disclosure : Total message number of messages for a discosure when the length of text is over the size 2. Index Constituents Information : Total number of contituents for an index
30607	CurrentMsgNo	Y	Int	16	1. Disclosure : Current message number of messages for a disclosure when the length of text is over the size 2. Index Constituents Information : Current sequence number of contituents for an index
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
	Standard Trailer	Y			

4.9 Random End

Distribute the information of Random End when the Random End is triggered. Random End is activated in every call auction session

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = MW
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
20004	BoardID	Y	String	2	Unique identifier to a Board where a given instrument is traded G1 : Main G2 : Pre open G3 : Post closing G4 : Odd lot G7 : Buy-in G8 : Sell-out T1 : Negotiation (regular) T4 : Negotiation (regular for Odd lot) T2 : Negotiation (pre) T3 : Negotiation (post) T6 : Negotiation (post for Odd lot) R1 : Negotiation (Repo) AL : All Boards
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
60	TransactTime	Y	UTCTime	9	Time of execution/order creation HHmmSSsss
30615	RandomEndApplyClassification	Y	String	1	Value of type code for applying or releasing random end 1: Opening call auction session RE set 2: Opening call auction session RE release 3: Closing single price auction session RE set 4: Closing single price auction session RE release 5: Post-market off-hour RE set 6: Post-market off-hour RE release
30616	RandomEndTentativeExecutionPrice	Y	Price	15(9.4)	Tentative price of the execution in the call auction when the random end is applied
30617	RandomEndEstimatedHighestPrice	Y	Price	15(9.4)	Estimated highest price in the call auction when the random end is applied
30618	RandomEndEstimatedHighestPriceDisparateRatio	Y	Float	13(5.6)	Disparate ration between estimated the highest price and tentative price in a call auction
30619	RandomEndEstimatedLowestPrice	Y	Price	15(9.4)	Estimated lowest price in the call auction when the random end is applied
30620	RandomEndEstimatedLowestPriceDisparateRatio	Y	Float	13(5.6)	Disparate ration between estimated the lowest price and tentative price in a call auction
30621	LatestPrice	Y	Price	15(9.4)	Price of the latest execution before the call auction
30622	LatestPriceDisparateRatio	Y	Float	13(5.6)	Disparate ration between the latest execution price and tentative price in a call auction

30623	RandomEndReleaseTime	N	String	9	Time of releasing the random end HHmmSSsss
	Standard Trailer	Y			

4.10 Price

This message is generated after every trade/or order is made, distribute the repeated structures of best orders in levels for best order messages, and distribute repeated structures of the execution/opening/highest/lowest price for trade messages.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = X
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
20004	BoardID	Y	String	2	Unique identifier to a Board where a given instrument is traded G1 : Main G2 : Pre open G3 : Post closing G4 : Odd lot G7 : Buy-in G8 : Sell-out T1 : Negotiation (regular) T4 : Negotiation (regular for Odd lot) T2 : Negotiation (pre) T3 : Negotiation (post) T6 : Negotiation (post for Odd lot) R1 : Negotiation (Repo) AL : All Boards
336	TradingSessionID	Y	String	2	Trading Session 01 = GT Order Load 10 = Opening Call Auction 11 = Opening Call Auction Extension 20 = Call Auction after Halt 21 = Call Auction after Halt Extension 30 = Closing Call Auction 40 = Continuous Auction 50 = VI Call Auction 51 = VI Call Auction Extension 60 = Order Accept Only 80 = Periodic Call Auction 90 = Trading Halt 99 = Board Closing
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want

75	TradeDate	Y	LocalMkt Date	8	Indicates local date of trade refereneced in this message in YYYYMMDD format. YYYYMMDD
60	TransactTime	Y	UTCTime	9	Time of execution/order creation HHmmSSsss
387	TotalVolumeTraded	N	Int	12	Total volume traded
381	GrossTradeAmt	N	Float	23(18.4)	Gross amount traded
30521	SellTotOrderQty	N	Int	12	Total quantity of remaining orders in ask side
30522	BuyTotOrderQty	N	Int	12	Total quantity of remaining orders in bid side
30523	SellValidOrderCnt	N	Int	11	Valid count of orders in ask side
30524	BuyValidOrderCnt	N	Int	11	Valid count of orders in bid side
268	NoMDEntries	Y	Int	9	Number(Depth) of market data entries repeated as following
Repeating Group					
83	RptSeq	Y	Int	9	Data sequential number within repeated record
279	MDUpdateAction	Y	String	1	Type code of update action for an entry of market data 0 = New
269	MDEntryType	Y	String	1	Data classification for an entry of market data 0 = Bid 1 = Offer 2 = Trade 4 = Opening Price 7 = Trading Session High Price 8 = Trading Session Low Price
290	MDEntryPositionNo	Y	Int	2	Position no (or level) for an entry of market data
270	MDEntryPx	Y	Price	15(9.4)	Price for an entry of market data
271	MDEntrySize	Y	Int	12	Size (or quantity) for an entry of market data
346	NumberOfOrders	N	Int	11	Number of orders
30270	MDEntryYield	N	Float	13(5.6)	Yield of the entry. Bond market only.
30271	MDEntryMMSize	N	Int	12	Size of the entry provided from market makers
	Standard Trailer	Y			

4.11 Price Recovery

This message is snapshot of market generated periodically containing deal and order information

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = W
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
20004	BoardID	Y	String	2	Unique identifier to a Board where a given instrument is traded G1 : Main

					G2 : Pre open G3 : Post closing G4 : Odd lot G7 : Buy-in G8 : Sell-out T1 : Negotiation (regular) T4 : Negotiation (regular for Odd lot) T2 : Negotiation (pre) T3 : Negotiation (post) T6 : Negotiation (post for Odd lot) R1 : Negotiation (Repo) AL : All Boards
336	TradingSessionID	Y	String	2	Trading Session 01 = GT Order Load 10 = Opening Call Auction 11 = Opening Call Auction Extension 20 = Call Auction after Halt 21 = Call Auction after Halt Extension 30 = Closing Call Auction 40 = Continuous Auction 50 = VI Call Auction 51 = VI Call Auction Extension 60 = Order Accept Only 80 = Periodic Call Auction 90 = Trading Halt 99 = Board Closing
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
30561	OpnPx	Y	Price	15(9.4)	The first price during that day
30562	TrdSessnHighPx	Y	Price	15(9.4)	Highest price during that day
30563	TrdSessnLowPx	Y	Price	15(9.4)	Lowest price during that day
20026	SymbolCloseInfoPx	Y	Price	15(9.4)	The last execution price of that day. If there is no execution it is replaced with the reference price, and if no reference price is given then it will be 0 (zero)
387	TotalVolumeTraded	Y	Int	12	Total volume traded
381	GrossTradeAmt	Y	Float	23(18.4)	Gross amount traded
30521	SellTotOrderQty	Y	Int	12	Total quantity of remaining orders in ask side
30522	BuyTotOrderQty	Y	Int	12	Total quantity of remaining orders in bid side
30523	SellValidOrderCnt	Y	Int	11	Valid count of orders in ask side
30524	BuyValidOrderCnt	Y	Int	11	Valid count of orders in bid side
268	NoMDEntries	Y	Int	9	Number(Depth) of market data entries repeated as following
Repeating Group					
83	RptSeq	Y	Int	9	Data sequential number within repeated record

279	MDUpdateAction	Y	String	1	Type code of update action for an entry of market data 0 = New
269	MDEntryType	Y	String	1	Data classification for an entry of market data 0 = Bid 1 = Offer 2 = Trade 4 = Opening Price 7 = Trading Session High Price 8 = Trading Session Low Price
290	MDEntryPositionNo	Y	Int	2	Position no (or level) for an entry of market data
270	MDEntryPx	Y	Price	15(9.4)	Price for an entry of market data
271	MDEntrySize	Y	Int	12	Size (or quantity) for an entry of market data
346	NumberOfOrders	Y	Int	11	Number of orders
30271	MDEntryMMSize	Y	Int	12	Size of the entry provided from market makers
	Standard Trailer	Y			

4.12 Price Recovery (Bond)

This message is snapshot of market generated periodically containing deal and order information

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = W
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
20004	BoardID	Y	String	2	Unique identifier to a Board where a given instrument is traded G1 : Main G2 : Pre open G3 : Post closing G4 : Odd lot G7 : Buy-in G8 : Sell-out T1 : Negotiation (regular) T4 : Negotiation (regular for Odd lot) T2 : Negotiation (pre) T3 : Negotiation (post) T6 : Negotiation (post for Odd lot) R1 : Negotiation (Repo) AL : All Boards

336	TradingSessionID	Y	String	2	Trading Session 01 = GT Order Load 10 = Opening Call Auction 11 = Opening Call Auction Extension 20 = Call Auction after Halt 21 = Call Auction after Halt Extension 30 = Closing Call Auction 40 = Continuous Auction 50 = VI Call Auction 51 = VI Call Auction Extension 60 = Order Accept Only 80 = Periodic Call Auction 90 = Trading Halt 99 = Board Closing
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
30561	OpnPx	Y	Price	15(9.4)	The first price during that day
30562	TrdSessnHighPx	Y	Price	15(9.4)	Highest price during that day
30563	TrdSessnLowPx	Y	Price	15(9.4)	Lowest price during that day
20026	SymbolCloseInfoPx	Y	Price	15(9.4)	The last execution price of that day. If there is no execution it is replaced with the reference price, and if no reference price is given then it will be 0 (zero)
30565	OpnPxYld	Y	Float	13(5.6)	Yield rate for opening price
30566	TrdSessnHighPxYld	Y	Float	13(5.6)	Yield rate for high price
30567	TrdSessnLowPxYld	Y	Float	13(5.6)	Yield rate for low price
30568	ClsPxYld	Y	Float	13(5.6)	Final yield of a bond calculated using the closing price of that day
387	TotalVolumeTraded	Y	Int	12	Total volume traded
381	GrossTradeAmt	Y	Float	23(18.4)	Gross amount traded
30521	SellTotOrderQty	Y	Int	12	Total quantity of remaining orders in ask side
30522	BuyTotOrderQty	Y	Int	12	Total quantity of remaining orders in bid side
30523	SellValidOrderCnt	Y	Int	11	Valid count of orders in ask side
30524	BuyValidOrderCnt	Y	Int	11	Valid count of orders in bid side
268	NoMDEntries	Y	Int	9	Number(Depth) of market data entries repeated as following
Repeating Group					
83	RptSeq	Y	Int	9	Data sequential number within repeated record
279	MDUpdateAction	Y	String	1	Type code of update action for an entry of market data 0 = New
269	MDEntryType	Y	String	1	Data classification for an entry of market data 0 = Bid 1 = Offer 2 = Trade 4 = Opening Price 7 = Trading Session High Price 8 = Trading Session Low Price
290	MDEntryPositionNo	Y	Int	2	Position no (or level) for an entry of market data
270	MDEntryPx	Y	Price	15(9.4)	Price for an entry of market data
271	MDEntrySize	Y	Int	12	Size (or quantity) for an entry of market data
346	NumberOfOrders	Y	Int	11	Number of orders

30270	MDEntryYield	Y	Float	13(5.6)	Yield of the entry. Bond market only.
30271	MDEntryMMSize	Y	Int	12	Size of the entry provided from market makers
	Standard Trailer	Y			

4.13 Index

Index message contains all real time data for a given index. This message is distributed periodically during trading time and EOD (from the end of the market to termination of process).

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = M1
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
336	TradingSessionID	Y	String	2	Trading Session 01 = GT Order Load 10 = Opening Call Auction 11 = Opening Call Auction Extension 20 = Call Auction after Halt 21 = Call Auction after Halt Extension 30 = Closing Call Auction 40 = Continuous Auction 50 = VI Call Auction 51 = VI Call Auction Extension 60 = Order Accept Only 80 = Periodic Call Auction 90 = Trading Halt 99 = Board Closing
30569	MarketIndexClass	Y	String	1	Market code of an index 1: HoSE 2: HNX 3: Composite (HoSE + HNX) 4. UPX
30167	IndexsTypeCode	Y	String	8	Industry Index Code for example 001: Composite index 002: VNALLshare index 100: VN30 index
15	Currency	Y	String	3	Currency code on ISO 4217 VND: Vietnamese Dong USD: US Dollar EUR: Euro
60	TransactTime	Y	UTCTime	9	Time of execution/order creation HHmmSSsss
30217	ValueIndexes	Y	Float	10(6.2)	Value of the index calculated
387	TotalVolumeTraded	Y	Int	12	Total volume traded
381	GrossTradeAmt	Y	Float	23(18.4)	Gross amount traded
30638	ContauctAccTrdvol	Y	Int	12	Accumulative trading volume of boards in continuous trading method
30639	ContauctAccTrdval	Y	Float	23(18.4)	Accumulative trading value of boards in continuous trading method
30640	BlktrdAccTrdvol	Y	Int	12	Accumulative trading volume of boards in block trading method

30641	BlktrdAccTrdval	Y	Float	23(18.4)	Accumulative trading value of boards in block trading method
30589	FluctuationUpperLimitIssueCount	Y	Int	7	Number of security of which price is upper limit
30590	FluctuationUpIssueCount	Y	Int	7	Number of security of which price is increasing
30591	FluctuationSteadinessIssueCount	Y	Int	7	Number of security of which price is not changed
30592	FluctuationDownIssueCount	Y	Int	7	Number of security of which price is decreasing
30593	FluctuationLowerLimitIssueCount	Y	Int	7	Number of security of which price is lower limit
30594	FluctuationUpIssueVolume	Y	Int	12	Sum of trading volumes of issues of which price is incresing or upper limit
30595	FluctuationDownIssueVolume	Y	Int	12	Sum of trading volumes of issues of which price is decresing or lower limit
30596	FluctuationSteadinessIssueVolume	Y	Int	12	Sum of trading volumes of issues of which price is not changed
	Standard Trailer				

4.14 Investor per Industry

This statistics message is generated periodically during trading hours providing sum of trade by Investor. The earliest data can be distributed after open call auction session.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = M2
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
60	TransactTime	Y	UTCTime	9	Time of execution/order creation HHmmSSsss
30569	MarketIndexClass	Y	String	1	Market code of an index 1: HoSE 2: HNX 3: Composite (HoSE + HNX) 4. UPX
30167	IndexsTypeCode	Y	String	8	Industry Index Code for example 001: Composite index 002: VNALLshare index 100: VN30 index
15	Currency	Y	String	3	Currency code on ISO 4217 VND: Vietnamese Dong USD: US Dollar EUR: Euro
20000	InvestCode	Y	String	4	This field indicates which category the owner of a given order or quote belongs to. This field is validated against AccountType field as below: - If AccountType is Customer (1), any value is permitted for this field. - If AccountType is House (3), this field must be 1000. 1000 = Securities company

					2000 = Insurance company 3000 = Asset management company and Investment company 3200 = Fund 4000 = Bank 5000 = General finance company 6000 = Pension Fund 7000 = Goverment representation 7030 = International organization 7100 = Misc corporation 8000 = Indivisual
331	SellVolume	Y	Int	12	Trading volume in sell side
30168	SellTradeAmount	Y	Float	23(18.4)	Trading value in ask side
330	BuyVolume	Y	Int	12	Trading volume in buy side
30169	BuyTradedAmount	Y	Float	23(18.4)	Trading value in bid side
	Standard Trailer	Y			

4.15 Investor per Industry (Bond)

This statistics message is generated periodically during trading hours providing sum of trade by Investor. The earliest data can be distributed after open call auction session.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = M2
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
60	TransactTime	Y	UTCTime	9	Time of execution/order creation HHmmSSsss
30597	BondClassificationCode	Y	String	10	T-bills, T-Bonds, Note of the central bank, State mortgage bonds and so on. ## If CFI has a function to classify above, this item is needless
20000	InvestCode	Y	String	4	This field indicates which category the owner of a given order or quote belongs to. This field is validated against AccountType field as below: - If AccountType is Customer (1), any value is permitted for this field. - If AccountType is House (3), this field must be 1000. 1000 = Securities company 2000 = Insurance company 3000 = Asset management company and Investment company 3200 = Fund 4000 = Bank 5000 = General finance company 6000 = Pension Fund

					7000 = Goverment representation 7030 = International organization 7100 = Misc corporation 8000 = Indivisual
331	SellVolume	Y	Int	12	Trading volume in sell side
30168	SellTradeAmount	Y	Float	23(18.4)	Trading value in ask side
330	BuyVolume	Y	Int	12	Trading volume in buy side
30169	BuyTradedAmount	Y	Float	23(18.4)	Trading value in bid side
	Standard Trailer	Y			

4.16 Investor per Industry (Derivatives)

This statistics message is generated periodically during trading hours providing sum of trade by Investor. The earliest data can be distributed after open call auction session.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = M2
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
60	TransactTime	Y	UTCTime	9	Time of execution/order creation HHmmSSsss
30604	SecurityGroupID	Y	String	2	Security group ID (Stock, Investment Trust, Futures, Option, etc.)
20000	InvestCode	Y	String	4	This field indicates which category the owner of a given order or quote belongs to. This field is validated against AccountType field as below: - If AccountType is Customer (1), any value is permitted for this field. - If AccountType is House (3), this field must be 1000. 1000 = Securities company 2000 = Insurance company 3000 = Asset management company and Investment company 3200 = Fund 4000 = Bank 5000 = General finance company 6000 = Pension Fund 7000 = Goverment representation 7030 = International organization 7100 = Misc corporation 8000 = Indivisual
331	SellVolume	Y	Int	12	Trading volume in sell side
30168	SellTradeAmount	Y	Float	23(18.4)	Trading value in ask side
330	BuyVolume	Y	Int	12	Trading volume in buy side
30169	BuyTradedAmount	Y	Float	23(18.4)	Trading value in bid side

	Standard Trailer	Y			

4.17 Investor per Symbol

This statistics message is generated at the end of day providing sum of trade by Investor for each symbol

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = M3
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
20000	InvestCode	Y	String	4	This field indicates which category the owner of a given order or quote belongs to. This field is validated against AccountType field as below: - If AccountType is Customer (1), any value is permitted for this field. - If AccountType is House (3), this field must be 1000. 1000 = Securities company 2000 = Insurance company 3000 = Asset management company and Investment company 3200 = Fund 4000 = Bank 5000 = General finance company 6000 = Pension Fund 7000 = Goverment representation 7030 = International organization 7100 = Misc corporation 8000 = Indivisual
331	SellVolume	Y	Int	12	Trading volume in sell side
30168	SellTradeAmount	Y	Float	23(18.4)	Trading value in ask side
330	BuyVolume	Y	Int	12	Trading volume in buy side
30169	BuyTradedAmount	Y	Float	23(18.4)	Trading value in bid side
	Standard Trailer	Y			

4.18 Top N Members per Symbol

This statistics message is generated periodically during trading hours providing top N members for each symbol. The earliest data can be distributed after open call auction session.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = M4

30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
911	TotNumReports	Y	Int	16	Entire number of messages for issue information transmission or for a set of statistics results
30215	SellRankSeq	Y	Int	6	Sequence of rank in sell side
30213	SellMemberNo	Y	String	5	Member ID in sell side
331	SellVolume	Y	Int	12	Trading volume in sell side
30168	SellTradeAmount	Y	Float	23(18.4)	Trading value in ask side
30216	BuyRankSeq	Y	Int	6	Sequence of rank in buy side
30214	BuyMemberNo	Y	String	5	Member ID in buy side
330	BuyVolume	Y	Int	12	Trading volume in buy side
30169	BuyTradedAmount	Y	Float	23(18.4)	Trading value in bid side
	Standard Trailer	Y			

4.19 Open Interest

This message is distributed at the start of a business date providing the total number of option contracts that are currently open

Tag	Field Name	Req'd	Format	Length	Comments
	Standard Header	Y			MsgType = MA
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
75	TradeDate	Y	LocalMktDate	8	Indicates local date of trade refereneced in this message in YYYYMMDD format. YYYYMMDD
30540	OpenInterestQty	Y	Number	10	Open interest quantity
30573	SettlementPrice	Y	Price	18(9.8)	Settlement price. It will be the reference price of tomorrow of a future instrument when it has any trade since its listing
	Standard Trailer	Y			

4.20 Deem Trade Price

This message provides expected trade price for a symbol during open call auction session.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = ME
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
20004	BoardID	Y	String	2	Unique identifier to a Board where a given instrument is traded G1 : Main G2 : Pre open G3 : Post closing G4 : Odd lot G7 : Buy-in G8 : Sell-out T1 : Negotiation (regular) T4 : Negotiation (regular for Odd lot) T2 : Negotiation (pre) T3 : Negotiation (post) T6 : Negotiation (post for Odd lot) R1 : Negotiation (Repo) AL : All Boards
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
30552	ExpectedTradePx	Y	Price	15(9.4)	Deemed trading price
30553	ExpectedTradeQty	Y	Int	12	
30554	ExpectedTradeYield	N	Float	13(5.6)	Yield of deemed trading price
	Standard Trailer	Y			

4.21 Foreigner Order Limit

This message is generated at the start of the business day providing the limited quantity to order by foreigners of each symbol, and updated whenever an order and/or a trade is made.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = MF
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market

					- UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
30557	ForeignerBuyPosblQty	Y	Int	15	Maximum number of shares allocated for foreign investors to buy. It is well known as "Total Room"
30558	ForeignerOrderLimitQty	Y	Int	16	Number of shares available for investors to buy at that time. It is well-known as “Current Room”. Current Room = Total Room – Foreigner Holding Shares
	Standard Trailer	Y			

4.22 Price Limit Expansion

This message is generated during trading hours whenever there is change of price limit of Derivatives product by market control.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = MX
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
20004	BoardID	Y	String	2	Unique identifier to a Board where a given instrument is traded G1 : Main G2 : Pre open G3 : Post closing G4 : Odd lot G7 : Buy-in G8 : Sell-out T1 : Negotiation (regular) T4 : Negotiation (regular for Odd lot) T2 : Negotiation (pre) T3 : Negotiation (post) T6 : Negotiation (post for Odd lot) R1 : Negotiation (Repo) AL : All Boards
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
60	TransactTime	Y	UTCTime	9	Time of execution/order creation HHmmSSsss

30649	PleUpLmtStep	N	Int	2	Final step number of PLE in the upper side actually triggered intraday
30650	PleLwLmtStep	N	Int	2	Final step number of PLE in the lower side actually triggered intraday
1149	HighLimitPrice	Y	Price	15(9.4)	Available high limit price for the trading day
1148	LowLimitPrice	Y	Price	15(9.4)	Available low limit price for the trading day
	Standard Trailer	Y			

4.23 ETF iNav

This message is generated periodically during trading hour providing the Indicative Net Asset value for each ETF symbol.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = MM
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
60	TransactTime	Y	UTCTime	9	Time of execution/order creation HHmmSSsss
30599	iNAVvalue	Y	Float	24(18.4)	Indicative Net Asset Value
	Standard Trailer	Y			

4.24 ETF iIndex

This message is generated periodically during trading hour providing the value of iIndex for each ETF symbol.
iIndex is calculated based on ETF's reference index (divided by a certain factor) to facilitate investors to compare the price movements of ETF fund certificates with reference index

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = MN
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want

60	TransactTime	Y	UTCTime	9	Time of execution/order creation HHmmSSsss
30217	ValueIndexes	Y	Float	10(6.2)	Value of the index calculated
	Standard Trailer	Y			

4.25 ETF Tracking Error

This message is distributed at the start of business day providing Tracking Error value calculated for each ETF symbol

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = MO
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
75	TradeDate	Y	LocalMktDate	8	Indicates local date of trade refereneced in this message in YYYYMMDD format. YYYYMMDD
30600	TrackingError	Y	Float	13(5.6)	Value of tracking error calculated
30602	DisparateRatio	Y	Float	24(16.6)	Value of disparate ratio for the tracking error which is calculated by closing price of the ETF - NAV
	Standard Trailer	Y			

4.26 Top N Symbols with Trading Quantity

This statistics message is generated periodically during trading hours providing Top N symbols with Trading Quantity. The earliest data can be distributed after open call auction session.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = MP
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
911	TotNumReports	Y	Int	16	Entire number of messages for issue information transmission or for a set of statistics results
30634	Rank	Y	Int	6	Sequential rank for 'Top N' statistics

55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
271	MDEntrySize	Y	Int	12	Size (or quantity) for an entry of market data
	Standard Trailer	Y			

4.27 Top N Symbols with Current Price

This statistics message is generated periodically during trading hours providing Top N symbols with Current price. The earliest data can be distributed after open call auction session.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = MQ
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
911	TotNumReports	Y	Int	16	Entire number of messages for issue information transmission or for a set of statistics results
30634	Rank	Y	Int	6	Sequential rank for 'Top N' statistics
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
270	MDEntryPx	Y	Price	15(9.4)	Price for an entry of market data
	Standard Trailer	Y			

4.28 Top N Symbols with High Ratio of Price

This statistics message is generated periodically during trading hours providing Top N symbols with High ratio of Price. The earliest data can be distributed after open call auction session.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = MR
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
911	TotNumReports	Y	Int	16	Entire number of messages for issue information transmission or for a set of statistics results
30634	Rank	Y	Int	6	Sequential rank for 'Top N' statistics

55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
30603	PriceFluctuationRatio	Y	Float	10(6.2)	Fluctuation ratio of the current price compared to the price of the previous day for a symbol
	Standard Trailer	Y			

4.29 Top N Symbols with Low Ratio of Price

This statistics message is generated periodically during trading hours providing Top N symbols with Low ratio of price. The earliest data can be distributed after open call auction session.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = MS
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
911	TotNumReports	Y	Int	16	Entire number of messages for issue information transmission or for a set of statistics results
30634	Rank	Y	Int	6	Sequential rank for 'Top N' statistics
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
30603	PriceFluctuationRatio	Y	Float	10(6.2)	Fluctuation ratio of the current price compared to the price of the previous day for a symbol
	Standard Trailer	Y			

4.30 Trading Result of Foreign Investors

This message distributes sum of trade by foreign investor for each symbol.

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Headerss	Y			MsgType = MT
30001	MarketID	Y	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market - RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
20004	BoardID	Y	String	2	Unique identifier to a Board where a given instrument is traded G1 : Main G2 : Pre open

					G3 : Post closing G4 : Odd lot G7 : Buy-in G8 : Sell-out T1 : Negotiation (regular) T4 : Negotiation (regular for Odd lot) T2 : Negotiation (pre) T3 : Negotiation (post) T6 : Negotiation (post for Odd lot) R1 : Negotiation (Repo) AL : All Boards
336	TradingSessionID	Y	String	2	Trading Session 01 = GT Order Load 10 = Opening Call Auction 11 = Opening Call Auction Extension 20 = Call Auction after Halt 21 = Call Auction after Halt Extension 30 = Closing Call Auction 40 = Continuous Auction 50 = VI Call Auction 51 = VI Call Auction Extension 60 = Order Accept Only 80 = Periodic Call Auction 90 = Trading Halt 99 = Board Closing
55	Symbol	Y	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
60	TransactTime	Y	UTCTime	9	Time of execution/order creation HHmmSSsss
20054	FornInvestTypeCode	Y	String	2	Indicates whether a given order or quote is owned by foreigner. 00 Domestic(default) 10 Foreginer
331	SellVolume	Y	Int	12	Trading volume in sell side
30168	SellTradeAmount	Y	Float	23(18.4)	Trading value in ask side
330	BuyVolume	Y	Int	12	Trading volume in buy side
30169	BuyTradedAmount	Y	Float	23(18.4)	Trading value in bid side
30643	SellVolume(Total)	Y	Int	12	Total trading volume in sell side
30644	SellTradeAmount(Total)	Y	Float	23(18.4)	Total trading value in ask side
30645	BuyVolume(Total)	Y	Int	12	Total trading volume in buy side
30646	BuyTradedAmount(Total)	Y	Float	23(18.4)	Total trading value in bid side
	Standard Trailer	Y			

4.31 Disclosure

This message distributes disclosure information from listing companies

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = MU
30001	MarketID	N	String	3	ID for identifying the market <Hochiminh Exchange> - STO: HoSE Stock Market - BDO: HoSE Bond Market

					- RPO: HoSE Repo Market <Hanoi Exchange> - STX: HNX Listed Stock Market - BDX: HNX Government Bond Market - DVX: HNX Derivative Market - UPX: HNX UpCoM Stock Market - HCX: HNX Corporate Bond Market
207	SecurityExchange	N	String	2	The unique ID assigned by Exchange HO: HoSE HX: HNX
55	Symbol	N	String	12	The unique code of an issue (a.k.a. instrument or symbol). Generally it is ISIN code but in case of Repo market it is defined as the exchange staffs want
30629	SymbolName	N	String	120	Instrument name
30605	DisclosureID	Y	String	16	Unique ID of a disclosure
30606	TotalMsgNo	Y	Int	16	1. Disclosure : Total message number of messages for a discosure when the length of text is over the size 2. Index Constituents Information : Total number of contituents for an index
30607	CurrentMsgNo	Y	Int	16	1. Disclosure : Current message number of messages for a disclosure when the length of text is over the size 2. Index Constituents Information : Current sequence number of contituents for an index
30608	LanguageCategory	Y	String	1	Value of type code for the language of the body text 1: Vietnamese 2: English
30609	DataCategory	Y	String	1	Whether the distributed data is on batch or in realtime 1: Batch 2: Realtime
30610	PublicInformationDate	Y	LocalMktDate	8	Date of the publication occured YYYYMMDD
30611	TransmissionDate	Y	LocalMktDate	8	Date of the transmission occured YYYYMMDD
30612	ProcessType	Y	String	1	Whether the distributed data is in normal or the revision version or to be deleted 1: Normal 2: Revision 3: Deleted
148	Headline	Y	String	250	TEXT Format
30613	Body	Y	String	1000	Body of disclosure
	Standard Trailer	Y			

4.32 Time Stamp (Polling)

Tag	Field Name	Req'd	Format	Length	Remarks
	Standard Header	Y			MsgType = MV
60	TransactTime	Y	UTCTime	9	Time of execution/order creation HHmmSSsss
	Standard Trailer	Y			

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